Neutron Stars as Fermi Gas

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I. INTRODUCTION

Stars maintain hydrostatic equilibrium for the majority of their lifespan as pressure from fusion balances gravity, but as fuel runs low and fusion processes slow, stellar cores shrink from the inbalance. For stars under eight solar masses ($< 8M_\odot$), the core shrinks until electron degeneracy pressure dominates, forming a white dwarf after the red giant phase, where Chandrasekhar famously found an upper mass limit of the resulting object ($\sim 1.4M_\odot$)[1]. The heaviest stars ($\geq 25M_\odot$) leave behind a black hole after going supernova from core collapse as the schwarzchild radius is crossed[2]. In the middle, we get some of the most extreme places in the universe, the neutron star. Neutron stars offer themselves to be a great observational tool as their special degenerate interiors are impossible to replicate on earth. These incredibly dense stellar objects were first theorized by Baade and Zwicky. They proposed that tightly packed neutrons in a stellar object would be responsible for the immense release of energy from observed supernova (a contender for the origin of cosmic rays).[3] The statistical mechanics was then heavily pioneered by Oppenheimer and Volkoff as they explored the structure of neutron stars by treating the interior as a cold and relativistic neutron gas under hydrostatic equilibrium in 1939.[4] Of course, we didn't get any neutron star observations until X-Ray astronomy kicked off in the 1960s and additional improvements to radio astronomy were made, where Giacconi indirectly observed extra-solar X-Rays and speculated sychrotron or stellar origins.[5] A few years later in 1967, an exciting discovery was made by Jocelyn Bell Burnell, who identified a rapid periodic signal while poring over pages of measurements. Potential intra-solar sources were crossed out and there was wonder of extra-terrestial intelligence, but similar pulsating sources identified across the sky solidifed that it was some natural phenomena. However, Bell Burnell's supervisor, Hewish, would be first author in the following publication, even after having pushed back on the statistical significance of the signal. He would then go on to recieve the Nobel Prize in Physics in 1974 after Gold and Pacini showed that these pulsating sources *were* rapidly rotating neutron stars, known as pulsars.[6,7] Bell Burnell's contribution would only be recognized years later when she was awarded the Special Breakthrough in Fundamental Physics award in 2018.[8]. Today, thousands of pulsars have been observed as recorded by the Australian Telescope National Facility Pulsar Catalogue.[9] The highly regular period of the pulsar pulse inspired Carl Sagan and Frank Drake to use this property to pinpoint earth on a pulsar map onboard

the voyager probe.[10] While electromagnetic-emitting neutron stars in the form of pulsars or even accreting neutron stars in X-Ray binaries are easier to detect, we will be focusing on the non-rotating case in our model.

As the name suggests, neutron stars are primarily composed of neutrons. They typically have radii of $R \sim 10\,\mathrm{km}$ and masses of about $M \sim 1 - 3M_\odot$[11]. The implications for the density are immense when considering that matter which is heavier the sun is being compressed into a space smaller than the width of the city of Calgary. This corresponds to an average density of about $\rho \sim 10^{15}\mathrm{g/cm}^3$. From here we must ask ourselves the central question of this paper. *What stops a neutron star from collapsing in on itself?* The answer will come from neutron degeneracy pressure associated with the pauli exclusion principle as it applies to a fermi gas.

II. QUANTUM FERMI GAS

Before we discuss neutron stars as Fermi gas, we must discuss Fermi gas. The formalism developed in this section serves two purposes: it provides the equation of state relating pressure to energy density in the neutron star interior, and it gives us the tools needed to assess whether the zero-temperature (cold gas) approximation is justified at the densities found within these objects.

A. Fermi–Dirac Statistics and the Density of States

Neutrons are spin-$\frac{1}{2}$ fermions, subject to the Pauli exclusion principle: no two neutrons may occupy the same quantum state simultaneously. In thermal and chemical equilibrium, the mean occupation number of a single-particle state with energy ϵ is given by the Fermi–Dirac distribution,

$$f(\epsilon) = \frac{1}{\exp[(\epsilon - \mu)/k_\mathrm{B}T] + 1}, \tag{1}$$

where μ is the chemical potential, k_B is Boltzmann's constant, and T is the temperature. For a gas of N neutrons in a volume V, counting momentum states with periodic boundary conditions and accounting for the two spin projections ($m_s = \pm\frac{1}{2}$), the number of states in the momentum interval $[p,\, p + dp]$ is

$$g(p)\, dp = \frac{8\pi V}{h^3}\, p^2\, dp, \tag{2}$$

where h is Planck's constant. This density of states is the fundamental object from which all thermodynamic quantities of the ideal Fermi gas are derived.

B. The Zero-Temperature Limit and the Fermi Energy

At zero temperature the Fermi-Dirac distribution (1) reduces to a step function: all states with $\epsilon < \mu(T=0) \equiv E_F$ are occupied, and all states above E_F are empty. The quantity E_F is the Fermi energy, and the corresponding momentum $p_F = \sqrt{2m_n E_F}$ (non-relativistically) is the Fermi momentum. Applying the number constraint where $N = \sum_k f(\epsilon)$ at $T = 0$ using (2) gives p_F in terms of n.

$$n \equiv \frac{N}{V} = \frac{8\pi}{h^3} \int_0^{p_F} p^2\, dp = \frac{8\pi}{3h^3}\, p_F^3, \tag{3}$$

$$p_F = \left(\frac{3h^3 n}{8\pi}\right)^{1/3}. \tag{4}$$

The corresponding Fermi temperature $T_F \equiv E_F/k_\mathrm{B}$ sets the scale below which quantum effects dominate. When $T \ll T_F$ the thermal smearing of the Fermi–Dirac distribution over an energy range $\sim k_\mathrm{B}T$ near E_F is negligible, and the zero-temperature approximation is excellent. We will check if this approximation is appropriate when talking about Neutron Star formulation.

C. Degeneracy Pressure

The Pauli exclusion principle forces neutrons to occupy states of progressively higher momentum as the density increases, contributing kinetic energy and hence pressure even at zero temperature and in the absence of inter-particle interactions. degeneracy pressure is what supports a neutron star against gravitational collapse.

The total kinetic energy of the gas at zero temperature is

$$E_\mathrm{kin} = \frac{8\pi V}{h^3} \int_0^{p_F} \mathcal{E}(p)\, p^2\, dp, \tag{5}$$

where $\mathcal{E}(p)$ is the single-particle energy. The pressure follows from the thermodynamic relation $P = -(\partial E_\mathrm{kin}/\partial V)_N$, yielding

$$P = \frac{8\pi c}{3h^3} \int_0^{p_F} \frac{p^4\, dp}{\mathcal{E}(p)/c}. \tag{6}$$

The specific form of $\mathcal{E}(p)$ determines which regime the gas occupies.

D. The Non-Relativistic Limit

When $p_F \ll m_n c$, the rest mass energy dominates and the single-particle energy is $\mathcal{E}(p) \approx m_n c^2 + p^2/2m_n$. Evaluating 6 in this limit gives

$$P_{non-rel} = \frac{1}{20}\left(\frac{3}{\pi}\right)^{2/3} \frac{h^2}{m_n}\, n^{5/3} \propto \rho^{5/3}, \tag{7}$$

where $\rho = m_n n$ is the mass density. This is a *polytropic* equation of state with index $\gamma = 5/3$, or equivalently polytropic index $\mathfrak{n} = 3/2$ in the Lane–Emden formalism that will be employed in Section IV.

E. The Ultra-Relativistic Limit

When $p_F \gg m_n c$, the rest mass is negligible and the single-particle energy approaches $\mathcal{E}(p) \approx pc$. Evaluating (6) in this limit gives

$$P_{rel} = \frac{hc}{8}\left(\frac{3}{\pi}\right)^{1/3} n^{4/3} \propto \rho^{4/3}. \tag{8}$$

The softening of the equation of state from $\gamma = 5/3$ to $\gamma = 4/3$ has profound consequences for stellar stability. A polytrope with $\gamma > 4/3$ can always find a new equilibrium if slightly compressed, since the pressure rises faster than gravity. When $\gamma = 4/3$ exactly, the star is marginally stable: no equilibrium exists above a critical mass, regardless of how much the central density increases. This is the physical origin of the maximum mass for compact stars, and it is the argument that underlies the Chandrasekhar limit for white dwarfs and, as shown in Section IV, a corresponding limit for neutron stars.

F. The Full Relativistic Equation of State

For neutron star interior conditions the Fermi momentum is neither much less nor much greater than $m_n c$, so the full relativistic dispersion relation $\mathcal{E}(p) = c\sqrt{p^2 + m_n^2 c^2}$ must be used. The energy density and pressure are then

$$\varepsilon = \frac{8\pi c}{h^3} \int_0^{p_F} \sqrt{p^2 + m_n^2 c^2}\, p^2\, dp, \tag{9}$$

$$P = \frac{8\pi c}{3h^3} \int_0^{p_F} \frac{p^4\, dp}{\sqrt{p^2 + m_n^2 c^2}}. \tag{10}$$

These integrals interpolate smoothly between the non-relativistic result (7) and the ultra-relativistic result (8) as $p_F/m_n c$ increases from 0 to ∞, and they constitute the equation of state that Oppenheimer and Volkoff used in their original calculation[4]. Closed-form expressions in terms of standard functions exist[1] but the integrals are straightforward to evaluate numerically.

III. NEUTRON STAR COMPOSITION AND FORMATION

With the thermodynamic properties of the ideal Fermi gas established in Section II, it is now possible to examine the physical conditions inside a neutron star and assess whether the free neutron Fermi gas is a justified model for the stellar interior. This section addresses three questions in turn: how neutron stars form, why their interiors are composed predominantly of neutrons, and why those neutrons constitute a highly degenerate Fermi gas to which the zero-temperature approximation applies.

A. Core-Collapse Supernovae and Neutron Star Formation

Neutron stars are the remnants of massive stars with main sequence masses $M \gtrsim 8\, M_\odot$ that end their lives in core-collapse supernovae[2]. Over the course of stellar evolution, successive stages of nuclear burning build up an iron core, which is the endpoint of exothermic fusion. When the iron core grows beyond the effective Chandrasekhar mass of approximately $1.4\, M_\odot$, electron degeneracy pressure can no longer support it and the core collapses. Collapse halts when the core reaches nuclear saturation density $\rho_0 \approx 2.3 \times 10^{14}$ g cm^{-3}, at which point the repulsive component of the nuclear force stiffens the equation of state and the core bounces, launching the supernova shock that disrupts the surrounding stellar envelope[12]. The remnant is a proto-neutron star that cools rapidly through neutrino emission, settling within seconds into a cold, degenerate object.

B. Neutronization and the Composition of the Interior

The composition of the neutron star interior is determined by the process of neutronization. In ordinary stellar matter, neutrons are unstable against beta decay $n \to p + e^- + \bar{\nu}_e$, with a free-neutron lifetime of approximately 15 minutes. However, at the extreme densities of the neutron star interior the reverse reaction,

$$p + e^- \to n + \nu_e, \tag{11}$$

becomes energetically favorable. This occurs when the electron chemical potential μ_e exceeds the neutron-proton mass difference,

$$\mu_e > (m_n - m_p)c^2 = \Delta m\, c^2 = 1.293 \text{ MeV}. \tag{12}$$

From equation (4) of Section II, the electron Fermi momentum rises with electron number density as $p_{F,e} \propto n_e^{1/3}$, and the corresponding ultra-relativistic Fermi energy $E_{F,e} = p_{F,e}c$ therefore increases with density. The condition (12) is satisfied once the density exceeds approximately 10^7 g cm^{-3}, well below the densities found throughout the bulk of a neutron star[11]. The result is that electron capture drives the composition toward a neutron-dominated state: in beta equilibrium, the proton fraction satisfies $Y_p \lesssim 0.1$ throughout most of the interior, and to a good first approximation the stellar matter may be treated as a single-component gas of free neutrons. It is this approximation that justifies the single-species Fermi gas model of Section II.

C. The Degeneracy Condition

The zero-temperature approximation adopted throughout Section II requires $T \ll T_F$, where the Fermi temperature $T_F = E_F/k_\mathrm{B}$ is set by the Fermi energy. Using equation (4), the non-relativistic Fermi energy of the neutron gas is

$$E_F = \frac{p_F^2}{2m_n} = \frac{\hbar^2}{2m_n}\left(3\pi^2 n\right)^{2/3}, \tag{13}$$

where $n = \rho/m_n$ is the neutron number density. Evaluating at a representative interior density $\rho \sim 2\rho_0 \approx 4.6 \times 10^{14}$ g cm^{-3} gives

$$E_F \sim 60 \text{ MeV}, \tag{14}$$

corresponding to a Fermi temperature

$$T_F = \frac{E_F}{k_{\rm B}} \sim 7 \times 10^{11}\ \text{K}. \tag{15}$$

The actual interior temperature of a neutron star older than a few minutes is $T \sim 10^8$–10^9 K, set by neutrino cooling following the proto-neutron star phase[13]. The ratio

$$\frac{T}{T_F} \sim 10^{-3}\text{–}10^{-2} \tag{16}$$

is small enough that thermal corrections to the equation of state are negligible at the level of accuracy of the free Fermi gas model. The zero-temperature approximation is therefore self-consistently justified, and the step-function form of the Fermi-Dirac distribution used in deriving equations (3)-(10) is an excellent approximation.

D. The Relativistic Parameter and Its Implications

A further check concerns the degree of relativistic degeneracy. The relevant dimensionless parameter is the ratio of the Fermi momentum to the neutron rest momentum $m_n c$,

$$x_F \equiv \frac{p_F}{m_n c} = \frac{\hbar}{m_n c}\left(3\pi^2 n\right)^{1/3}. \tag{17}$$

Evaluating at $\rho \sim \rho_0$ gives $x_F \sim 0.35$, rising to $x_F \sim 0.7$ at $\rho \sim 2\rho_0$. Since x_F is neither much less than nor much greater than unity, the neutrons occupy the *intermediate relativistic regime*: neither the non-relativistic limit $P \propto \rho^{5/3}$ nor the ultra-relativistic limit $P \propto \rho^{4/3}$ is uniformly valid throughout the star. This confirms that the full relativistic equation of state, equations (9)–(10), is necessary for a quantitative treatment, as employed by Oppenheimer and Volkoff[4].

E. Internal Structure and the Domain of the Model

Neutron stars are not uniform objects: they possess a layered internal structure whose composition varies with depth[14]. The outer crust consists of a lattice of neutron-rich nuclei in a relativistic electron gas, the inner crust contains free neutrons coexisting with nuclear clusters, and the inner core at densities above $\sim 2\rho_0$ may harbor exotic phases whose equation of state is poorly constrained. The free neutron Fermi gas is a reasonable approximation

only in the outer core, spanning roughly $0.5\rho_0 \lesssim \rho \lesssim 2\rho_0$. This region constitutes the bulk of the stellar volume and contains most of the mass, so the model captures the dominant physics governing the global properties of the star even though it is not valid everywhere.

IV. STELLAR STRUCTURE: NEWTONIAN HYDROSTATIC EQUILIBRIUM

The equation of state derived in Section II relates the pressure of the neutron gas to its energy density, but it says nothing by itself about how the star is structured. To determine the density profile $\rho(r)$, pressure profile $P(r)$, and the relationship between total mass M and radius R, a second condition is required: hydrostatic equilibrium. In this section that condition is formulated in the Newtonian framework, which provides physical transparency and a natural first pass at the mass-radius relationship before the full general relativistic treatment of Section V.

A. Hydrostatic Equilibrium

Consider a spherically symmetric star in static mechanical equilibrium. A thin shell of matter at radius r with thickness dr and density $\rho(r)$ is acted on by two forces: the pressure difference across the shell, which acts outward, and the gravitational attraction of all the mass $M(r)$ interior to r, which acts inward. Balancing these forces per unit area gives the condition of hydrostatic equilibrium,

$$\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}, \tag{18}$$

supplemented by the mass continuity equation,

$$\frac{dM}{dr} = 4\pi r^2 \rho(r). \tag{19}$$

Equations (18) and (19) together with the equation of state $P = P(\rho)$ from Section II form a closed system of ordinary differential equations. Given a central density $\rho_c \equiv \rho(0)$ as the single free parameter, integrating outward from $r = 0$ with initial conditions $M(0) = 0$ and $P(0) = P(\rho_c)$ until $P(R) = 0$ defines the stellar surface and yields the total mass $M \equiv M(R)$ and radius R. Varying ρ_c then traces out the full mass-radius relationship.

B. The Polytropic Equation of State and the Lane-Emden Equation

In the non-relativistic limit the equation of state takes the polytropic form $P = K\rho^\gamma$ with $\gamma = 5/3$, where the constant K follows from equation (7),

$$K = \frac{1}{20}\left(\frac{3}{\pi}\right)^{2/3}\frac{h^2}{m_n^{8/3}}. \tag{20}$$

A polytrope with $\gamma = 5/3$ corresponds to polytropic index $\mathfrak{n} = 1/(\gamma - 1) = 3/2$. Applying the standard Lane-Emden substitution[15], the hydrostatic equilibrium and mass continuity equations reduce to the Lane-Emden equation,

$$\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\theta}{d\xi}\right) = -\theta^{\mathfrak{n}}, \tag{21}$$

with boundary conditions $\theta(0) = 1$ and $\theta'(0) = 0$, where ξ is a dimensionless radial coordinate and θ is related to the density by $\rho = \rho_c\,\theta^{\mathfrak{n}}$. The stellar surface is located at the first zero ξ_1 of θ. For $\mathfrak{n} = 3/2$ the equation has no closed-form solution but is straightforward to integrate numerically, yielding $\xi_1 \approx 3.654$ and $|\theta'(\xi_1)| \approx 0.2033$[15].

C. The Mass-Radius Relation in the Non-Relativistic Limit

Eliminating the central density ρ_c between the Lane–Emden expressions for the total mass and radius yields the mass-radius relation for the non-relativistic neutron Fermi gas,

$$R \propto M^{-1/3}. \tag{22}$$

This inverse relationship is physically significant: unlike ordinary stars, which expand as mass is added, a degenerate star shrinks as its mass increases. Adding mass raises the central density and hence the Fermi momentum and degeneracy pressure, but the increased gravity compresses the star to a smaller radius. Evaluating at $M = 1.4\,M_\odot$ gives $R \sim 10$–15 km, consistent in order of magnitude with observed neutron star radii.

D. The Maximum Mass and the Breakdown of the Non-Relativistic Treatment

The mass-radius relation (22) has no upper bound on M in the strict non-relativistic limit: the star simply becomes smaller and denser without limit. However, as the central density

increases, $p_F/m_n c$ grows and the equation of state softens toward the ultra-relativistic limit $P \propto \rho^{4/3}$, corresponding to a polytropic index $\mathfrak{n} = 3$. The Lane–Emden equation for $\mathfrak{n} = 3$ has a qualitatively different character: the total mass,

$$M_{\rm Ch} = 4\pi\alpha^3\rho_c \left(-\xi^2\theta'\right)_{\xi_1} \propto K^{3/2}G^{-3/2}, \tag{23}$$

becomes independent of the central density ρ_c. This means that for a $\gamma = 4/3$ polytrope there is a unique mass for which equilibrium is possible, and no equilibrium exists above it regardless of how high the central density is driven. This is the *Chandrasekhar mass limit*, originally derived for white dwarfs[16] but applicable here with the electron mass replaced by the neutron mass. Substituting the neutron Fermi gas constant K from the ultra-relativistic equation of state (8) gives a Chandrasekhar-type limit for neutron stars of

$$M_{\rm Ch}^{(n)} \approx 5.73 \left(\frac{\hbar c}{G}\right)^{3/2} \frac{1}{m_n^2} \approx 5.4\, M_\odot. \tag{24}$$

This estimate is considerably larger than the actual maximum neutron star mass because it neglects two important effects that both act to reduce the maximum mass: the transition from Newtonian to general relativistic gravity, and the fact that the equation of state is not purely ultra-relativistic at the densities where the maximum mass is reached. Both effects are incorporated in the TOV treatment of Section V, which reduces the maximum mass to the Oppenheimer-Volkoff limit of $\approx 0.7\, M_\odot$.

The Newtonian analysis therefore identifies the correct physical mechanism. The softening of the equation of state from $\gamma = 5/3$ toward $\gamma = 4/3$ as relativistic effects become important, but overestimates the maximum mass by nearly an order of magnitude. Two approximations must be relaxed simultaneously to obtain a quantitatively reliable result: the non-relativistic equation of state must be replaced by the full relativistic form of equations (9) and (10), and Newtonian gravity must be replaced by the general relativistic equations of stellar structure. This is the task of Section V.

V. GENERAL RELATIVISTIC STELLAR STRUCTURE AND THE TOV EQUATION

The Newtonian analysis of Section IV identified the correct physical mechanism underlying the neutron star maximum mass but overestimated it by nearly an order of magnitude.

Two approximations must be relaxed to obtain a quantitatively reliable result: the equation of state must be taken in its full relativistic form, and the gravitational field must be treated within general relativity rather than Newtonian theory. The necessity of the latter is quantified by the compactness parameter,

$$\mathcal{C} \equiv \frac{GM}{Rc^2} \sim \frac{G \cdot 1.4\, M_\odot}{(10\ \mathrm{km})\, c^2} \sim 0.2, \tag{25}$$

which measures the ratio of the Schwarzschild radius $r_s = 2GM/c^2$ to the stellar diameter. For the Sun $\mathcal{C} \sim 10^{-6}$, and Newtonian gravity is an excellent approximation. For a neutron star $\mathcal{C} \sim 0.2$, placing it firmly in the regime where general relativistic corrections are of order 20% and cannot be neglected. This section derives the Tolman–Oppenheimer–Volkoff (TOV) equation, the general relativistic generalization of hydrostatic equilibrium, and uses it together with the equation of state of Section II to determine the maximum stable neutron star mass.

A. The Schwarzschild Metric and Stellar Structure

For a static, spherically symmetric mass distribution the spacetime geometry is described by the Schwarzschild-like interior metric,

$$ds^2 = -e^{2\Phi(r)} c^2\, dt^2 + \left(1 - \frac{2Gm(r)}{rc^2}\right)^{-1} dr^2 + r^2\, d\Omega^2, \tag{26}$$

where $\Phi(r)$ is the gravitational potential and $m(r)$ is the gravitational mass enclosed within radius r[17]. The mass function $m(r)$ satisfies the same continuity equation as in the Newtonian case,

$$\frac{dm}{dr} = 4\pi r^2 \varepsilon(r)/c^2, \tag{27}$$

except that the source term is now the total energy density ε (including rest mass) rather than the rest mass density ρ alone, reflecting the fact that in general relativity all forms of energy gravitate.

B. Derivation of the TOV Equation

The equation of hydrostatic equilibrium in general relativity is obtained from the covariant conservation of the stress-energy tensor, $\nabla_\mu T^{\mu\nu} = 0$, applied to a perfect fluid with

energy density ε and isotropic pressure P[4,18]. The r-component of this conservation law, evaluated using the metric (26), yields the Tolman–Oppenheimer–Volkoff equation,

$$\frac{dP}{dr} = -\frac{G}{r^2}\left(\varepsilon + \frac{P}{c^2}\right)\left(m(r) + \frac{4\pi r^3 P}{c^2}\right)\left(1 - \frac{2Gm(r)}{rc^2}\right)^{-1}, \tag{28}$$

which recovers the Newtonian equation (18) in the limit $c \to \infty$. The TOV equation, the mass continuity equation (27), and the equation of state $P = P(\varepsilon)$ from Section II form a closed system of ordinary differential equations in the fully general relativistic setting.

C. Numerical Integration and Boundary Conditions

The TOV system is integrated numerically outward from $r = 0$ with boundary conditions $m(0) = 0$ and $P(0) = P(\varepsilon_c)$, where ε_c is the chosen central energy density. Integration proceeds until $P(R) = 0$, which defines the stellar surface at radius R, and the total gravitational mass is $M = m(R)$. The equation of state is evaluated numerically from equations (9) and (10), with the Fermi momentum p_F as the parameter relating ε and P implicitly through equation (3). Repeating the integration for a sequence of central densities ε_c traces out the full mass-radius curve.

D. The Oppenheimer–Volkoff Maximum Mass

The central result of the TOV integration with the free neutron Fermi gas equation of state is the existence of a maximum stable mass. As the central density increases, the mass $M(\varepsilon_c)$ rises to a maximum and then decreases; configurations on the decreasing branch are unstable to radial perturbations[11]. The maximum mass obtained from this calculation is the Oppenheimer-Volkoff limit,

$$M_{\rm OV} \approx 0.7\, M_\odot, \tag{29}$$

occurring at a central density $\varepsilon_c \approx 5 \times 10^{15}$ g cm^{-3} and a corresponding radius $R \approx 9.6$ km[4]. This is the neutron star analog of the Chandrasekhar limit for white dwarfs, and it marks the boundary between stable neutron stars and objects that must collapse further — in the absence of additional pressure support — to black holes.

Comparing equations (24) and (29), the inclusion of general relativity reduces the maximum mass from the Newtonian estimate of $\sim 5.4\, M_\odot$ to $0.7\, M_\odot$, a reduction by nearly an

order of magnitude. This dramatic difference reflects the cumulative effect of the three GR correction terms identified in Section 5.3: each individually increases the effective gravitational pull, and together they render the star unstable at a far lower mass than Newtonian gravity would predict. The physical content of the OV limit is therefore not just a quantitative correction to the Chandrasekhar result but a qualitatively different regime in which spacetime curvature plays a central role in determining stellar stability.

VI. THE MASS–RADIUS RELATIONSHIP

The TOV integration of Section V, carried out across a range of central energy densities ε_c with the free neutron Fermi gas equation of state, produces a continuous curve in the mass-radius plane. This mass-radius (M–R) curve is the primary observable prediction of the model and the natural point at which theory meets observation.

A. Structure of the Mass–Radius Curve

The M–R curve for the free neutron Fermi gas has a characteristic shape that reflects the interplay between the equation of state and gravity as the central density is varied. Beginning at low central densities, both M and R are small. As ε_c increases, the mass rises steeply while the radius varies more slowly, broadly consistent with the Newtonian scaling $R \propto M^{-1/3}$ derived in Section IV: the star becomes more massive and more compact simultaneously. The curve reaches a maximum mass at $M_{\rm OV} \approx 0.7\, M_\odot$, corresponding to a radius of $R \approx 9.6$ km and a central density $\varepsilon_c \approx 5 \times 10^{15}$ g cm^{-3}[4]. Beyond this point, further increases in central density yield configurations of decreasing mass and decreasing radius: these are unstable to radial perturbations and would collapse to black holes if perturbed[11]. Only the branch of the M–R curve with $dM/d\varepsilon_c > 0$ corresponds to stable stellar configurations.

The termination of the stable branch at $M_{\rm OV}$ is a direct consequence of the softening of the equation of state discussed in Section V: as the central density approaches and exceeds $\varepsilon_c^{\rm max}$, the neutrons become increasingly ultra-relativistic and the equation of state asymptotes toward $P \propto \varepsilon^{4/3}$, at which point the three general relativistic correction terms in the TOV equation collectively overwhelm the pressure support.

B. Comparison with Observed Neutron Stars

The free Fermi gas model can be assessed quantitatively by comparing its M–R predictions against well-measured neutron stars. Table I lists a representative selection of neutron stars with precise mass and/or radius measurements drawn from pulsar timing and X-ray observations.

TABLE I. Selected neutron stars with well-measured masses and radii, compared to the Oppenheimer–Volkoff limit for the free neutron Fermi gas. Radii are from NICER X-ray timing.

Object	Mass ($M_\odot$)	Radius (km)	Reference
PSR J0740+6620†	2.08 ± 0.07	$12.39^{+1.30}_{-0.98}$	19,20
PSR J0030+0451	$1.34^{+0.15}_{-0.16}$	$12.71^{+1.14}_{-1.19}$	21
PSR J0437-4715	1.418 ± 0.037	$11.36^{+0.95}_{-0.63}$	22
PSR J1614-2230†	1.908 ± 0.016	—	23
OV limit (free Fermi gas)	≈ 0.70	≈ 9.6	4

Several features of Table I are immediately striking. First, every observed neutron star has a mass that exceeds $M_{\rm OV} \approx 0.7\,M_\odot$, with the most massive known example (from this list), PSR J0740+6620, reaching $2.08\,M_\odot$,nearly three times the OV limit. Second, the observed radii cluster around 11–13 km, systematically larger than the ≈ 9.6 km radius at the OV maximum. Both discrepancies point in the same direction: the true equation of state is stiffer than the free Fermi gas prediction, meaning that at a given energy density the actual pressure exceeds what a non-interacting neutron gas would produce. The physical origin of this is the repulsive component of the nuclear force at supranuclear densities, which is absent from the free Fermi gas model by construction.[24]

VII. CONCLUSION

This paper has developed the free neutron Fermi gas model of neutron star structure from first principles, following the framework established by Oppenheimer and Volkoff[4]. Beginning from the Fermi-Dirac distribution and the density of states in momentum space, the equation of state of a cold, ideal Fermi gas was derived in both the non-relativistic and

ultra-relativistic limits, as well as the full relativistic form. Application of this equation of state to the problem of stellar structure. This done first in the Newtonian framework via the Lane-Emden equation, then in full general relativity via the TOV equation where we yielded the mass-radius relationship and the Oppenheimer-Volkoff maximum mass of $M_{\mathrm{OV}} \approx 0.7\, M_{\odot}$. The macroscopic properties of neutron stars, their mass, radius, and the existence of a maximum mass, emerge as direct consequences of the Pauli exclusion principle applied at nuclear densities.

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